

Hyper-Threading allows multi-threaded server applications to execute multiple threads at once, and in parallel on a single-processor core.

PAT stands for Performance Acceleration Technology, a technique Intel introduced with their 875 Chipset, wherein the system performance is improved by optimising memory access between the CPU and system memory.

PCI stands for Peripheral Connect Interface, a standard for connecting various expansion cards in a micro-computer system. It succeeded the ISA standard used in older computers. This interconnecting bus is 32 bits wide, and runs at 33 MHz.

PCI-X stands for PCI-extended; the

evolution of the PCI bus saw the bus speed increasing to 66 MHz and the width extending to 64 bits. PCI-X is often used for connecting high-bandwidth requirement devices such as RAID cards and SCSI controllers.

PCI-E stands for PCI-Express, a new standard that promises to replace the older PCI standard due to its high bandwidth and other advantages. Each device connected on this bus will have four times the bandwidth offered by a regular PCI bus.

RAID stands for Redundant Array of Inexpensive Disks. The technique behind RAID is to use small, inexpensive disks to build an array of disks, which provide fault tolerance by making redundant copies of

data. There are various architectures, from RAID 1 to RAID 5.

SCSI stands for Small Computer System Interface. This is an interface that allows for the connecting of internal or external peripheral devices via a SCSI card, which can connect up to 15 devices per channel. It can reach a peak speed of 320 megabytes per second. There are various types of SCSI, classified on speed, such as Ultra 320, Ultra 160, etc.

SATA stands for Serial ATA, the new standard that is soon to replace the PATA interface standard, often used for connecting devices such as hard drives and optical drives. Being serial, it is a point-to-point interface, and hence offers higher bandwidth.

ported by three fans. Component accessibility is a non-issue due to abundant space. A screwless design ensures tool-free maintenance and installation of drives.

HP's ProLiant comes within a regular ATX cabinet. However, air flow is maintained well via a large 120 mm fan. Additionally, the internal wiring is clamped properly, thus avoiding clutter. The ProLiant ML110 is a showcase for tool-free design, easy accessibility to components, etc. Chassis intrusion detection was, however, missing.

Acer, too, had tool-free servers, clamped wiring, and drives that can be removed and re-attached without fuss. The server can be locked to avoid unrestricted access, or to prevent people from turning it off. A chassis intrusion-detection system is also present.

PCS does a great job on the ventilation front, with the help of deflectors that direct air-flow.

Zenith's Data Server was the worst-designed: the wires were not clamped properly and dangled dangerously close to the CPU fan. The front was completely packed with little room for air to be sucked in. The box is a typical desktop box, and an administrator would require effort to perform everyday maintenance.

Performance: Who Fared How

We did a two-tier performance test on these servers. First, we ran the usual benchmark suites to determine component-level performance. Then tests such as NetBench, WebBench and Oratools were used to evaluate the server's overall performance.

We take a look at how each server fared in the component-level tests. In the PovRay and ZD Bench Content Creation 2004, PCS's server topped. Following closely was Zenith's Data



The Acer G520: The best when it comes to performance and scalability

To sum it up, if you are looking for an all-rounder, nothing beats the Acer Altos G520. It is also the best from a scalability point of view

Server. These results were significantly better than those of other servers. This was quite interesting since there were better chipsets around, and some good configurations too, but the PCS server managed to beat the rest in Content Creation 2004. Also, this was the only server to have a GeForce class graphics card, and hence would make a good choice as a workstation.

In the SiSoft Sandra memory subsystem test, the variation was striking. Wipro and Zenith posted the best memory scores, followed by Acer's Altos G520. Both Wipro and Zenith had 400 MHz memory configured in dual-channel mode, and therefore scored well. The Altos G520 runs on a DDR 333 MHz sub-system, which was reflected in its score. All the other machines were running 400 MHz RAM but were configured in single-channel mode, effectively halving their potential bandwidth. This clearly shows the boost a server gets when memory is configured in dual channel mode.

The processor subsystem revealed nothing much except for the fact that Zenith's 3 GHz processor was beating the rest, which were running on 2.8 GHz processors.

In the hard disk sub-system, nothing came

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